

Advanced Public Economics — Applied Tutorial

Session V: Microsimulation Models I

Daniel Weishaar

Roadmap for Session V

Use a microsimulation model for the United States to ...

- ① ... get a more realistic picture of **effective marginal tax rates (EMTR)**,
- ② ... understand different **components of EMTR**.

Microsimulation Models — Overview

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- ① ... estimate tax liabilities and effective marginal tax rates (EMTR) very precisely for households with very specific characteristics.
- ② ... usually incorporate different years and geographic levels (federal, state,...).
- ③ ... (can be extended to simulate behavioral responses to taxes at the household level, e.g., using a discrete choice architecture, and to evaluate consequences of reforms *ex-ante*).

Microsimulation Models — Examples of Different Models

Country-Specific MSM

① United States:

- ▶ [Policy engine](#): open source based on [openfisca platform](#).
- ▶ [NBER TAXSIM](#): remote server program, different access options ⇒ **Today**.

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Cross-Country MSM

- ▶ [EUROMOD](#): Tax-benefit simulation for the EU countries. GUI access.
- ▶ [OECD TaxBen](#): Tax-benefit simulation for the OECD countries. Online access.

NBER TAXSIM — Overview

Coverage: US federal (and state) income tax laws, 1960-2023.

Inputs: Up to 35 inputs ranging from tax system information (year, state), demographics (marital status, age, ...) , to income information (wages, self-employment income, interest income...).

Outputs: Federal and state income taxes, marginal tax rates, tax credits, ...

Data: TAXSIM can be used with hypothetical households, but also with restricted access tax return data on the NBER server (see, e.g., Bierbrauer et al., 2021), or survey data like the Current Population Survey (see, e.g., Bierbrauer et al., 2023).

Access Options: server computation can be accessed via [online interface](#) and different programming languages (Stata, R, Julia).

- ▶ Online interface useful if you are interested in a handful of cases.
- ▶ We use R to send multiple cases directly to TAXSIM server ([low level interface](#))

NBER TAXSIM — Low-Level Interface — Concept

Key idea: TAXSIM is a program running on a remote server that communicates via **standard input** and **standard output**.

- ① The user prepares a **CSV file** with inputs.
- ② This file is sent to the TAXSIM server via an **SSH connection**.
- ③ TAXSIM reads the data and computes output (taxes, mtr, ...).
- ④ The output file is sent back as a CSV file.

Check whether SSH is enabled/Installed: `ssh_path <- Sys.which("ssh")`

⇒ If SSH is not installed, follow [instructions](#).

NBER TAXSIM — Low-Level Interface — Prepare Input File

TAXSIM requires csv input file with different components:

- a. **Tax system:** *year, taxsimid*
- b. **Household characteristics:** *mstat, page, sage, pwages, swages,...*, see [full list](#).
- c. **Controls of TAXSIM**
 - ▶ **idtl:** standard output (0), full output (2), text description (5)
 - ▶ **mtr:** with respect to wages (11), long term capital gains (70), primary wage earner (85), secondary wage earner (86).

⇒ Put all this information in one data frame and save as csv file.

NBER TAXSIM — Low-Level Interface — Send CSV File via SSH

Robust cross-platform call within R:

```
status <- system2(  
  command = "ssh",  
  args = c(  
    "-T",                                # no pseudo-terminal (non-interactive)  
    "-o", "StrictHostKeyChecking=no",    # don't ask to trust the host  
    "-o", "UserKnownHostsFile=/dev/null", # don't store host keys  
    "-p", "22",                          # default SSH port  
    "taxsim35@taxsimssh.nber.org"        # TAXSIM SSH server (user@host)  
  ),  
  stdin  = infile,                       # Prespecified csv input file paths  
  stdout = outfile                       # Prespecified csv output file paths  
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Debugging: If you get connection errors, try `-p 443` (or `-p 80`).

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⇒ After receiving results, output is stored in CSV file for import back into R.

NBER TAXSIM — Example

Liam and Charlotte Jones live in Kentucky. They are married and file a joint tax return for the tax year 2000. Liam is 45 years old and earns 75,000 USD in wages, Charlotte is 42 years old and earns wages of 0 USD. They have two children, Anna (16 years) and Peter (13 years).

⇒ **Hands-on exercise:** Compute federal and state taxes, as well as corresponding MTR with respect to wages.

NBER TAXSIM — Example — Outputs

Main Outputs:

Variable Name	Description	Value
fiitax	Federal income tax liability	9,106 USD
siitax	State income tax liability	4,016 USD
frate	Federal marginal tax rate	28%
srate	State marginal tax rate	6%

Additional Output Components visible here, e.g., social security and medicare withholding, tax credit information, ... (see full list and description [here](#)).

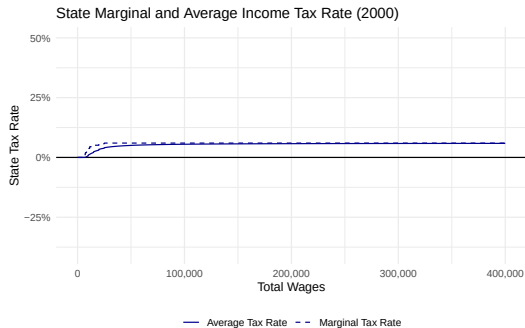
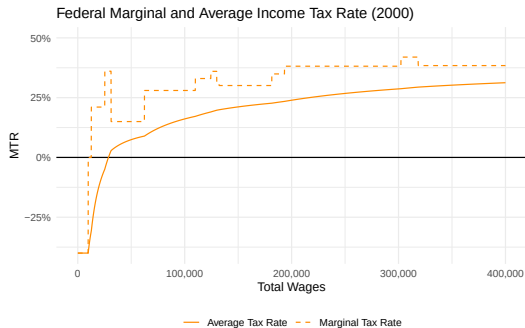
Note: By setting `idt1=5`, you receive the full text output as in the [online interface](#). More comprehensive, but less structured.

NBER TAXSIM — Analyze MTR and ATR Schedule

What does the EMTR schedule of Liam and Charlotte look like if Liams earnings are varied from 0 to 400,000 USD? $\Rightarrow$ [Hands-on exercise](#).

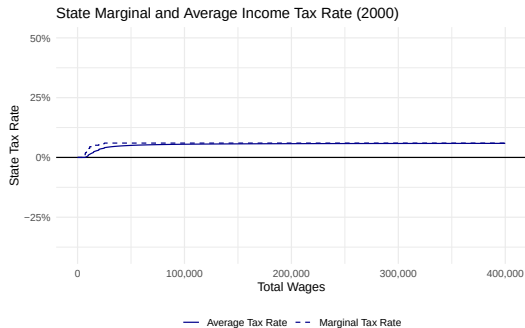
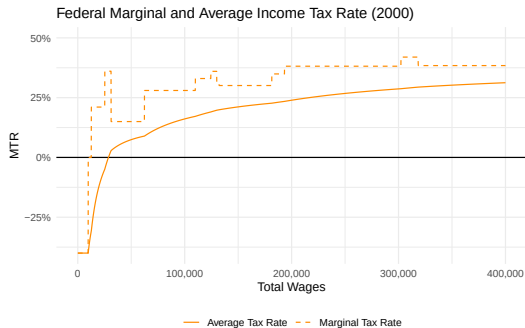
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Observations: Negative MTR/ATR at bottom, upward and downward MTR jumps.

NBER TAXSIM — Analyze Federal Taxes

Let's focus on the federal tax liability and marginal tax rate.

- ▶ To understand federal EMTR better, provide **decomposition** of tax liability.
- ▶ **Careful!** For different years and household types, decomposition might vary (different programs at different times).
- ▶ In 2000, there are three relevant components for our example household:

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⇒ **Hands-on exercise:** Visualize decomposition and compute MTR of components.

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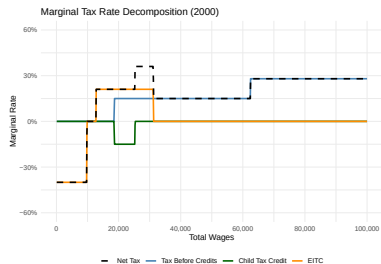
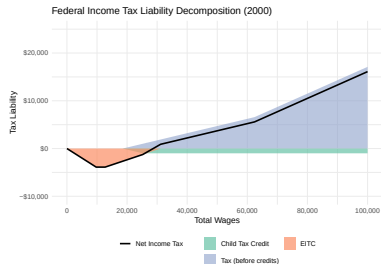
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- ▶ In 2000, there are three relevant components for our example household:

$$\underbrace{\text{net tax liability}}_{fiitax} = \underbrace{\text{tax before credits}}_{v28} - \underbrace{\text{earned income tax credit}}_{v25} - \underbrace{\text{child tax credit}}_{v22}$$

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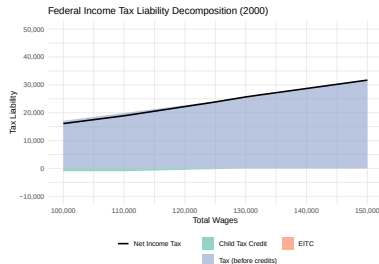
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Observations:

- ▶ EITC is phased in ($MTR_{EITC} < 0$), then held constant ($MTR_{EITC} = 0$), and then phased out ($MTR_{EITC} > 0$).
- ▶ Once EITC is phased out completely, $MTR_{EITC} = 0$. Responsible for drop in EMTR.

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- ▶ Child Tax Credit is phased out between 110,000 USD and 130,000 USD. Yields another drop.

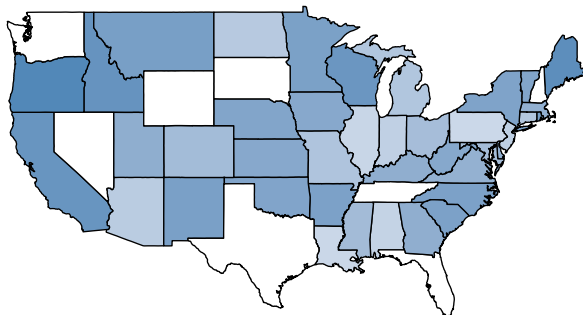
NBER TAXSIM — Analyze State MTR

What does the state MTR look like if Liam and Charlotte live in different states? $\Rightarrow$
[Hands-on exercise](#).

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State Marginal Tax Rate (MTR) for Example Household (2000)



State MTR
0.0% 2.5% 5.0% 7.5%

Observations: State MTR for Liam and Charlotte in year 2000 varies strongly, lies between 0 (e.g., Alaska, Florida, Nevada, New Hampshire, South Dakota, Tennessee, Texas, Washington, Wyoming) and 9 percent (District of Columbia).

That's it! See you next time!

References

References I

-  BIERBRAUER, F. J., P. BOYER, A. PEICHL, and D. WEISHAAR (2023). “The taxation of couples.” CEPR Discussion Paper 18138.
-  BIERBRAUER, F. J., P. C. BOYER, and A. PEICHL (2021). “Politically Feasible Reforms of Nonlinear Tax Systems.” *American Economic Review* 111 (1), pp. 153–191.

Instructions to Install SSH

macOS

- ▶ SSH is installed by default on almost all Macs
- ▶ If missing, macOS will prompt to install Command Line Tools
- ▶ Manual install (Terminal): `xcode-select --install`

Windows 10 / Windows 11

- ▶ Go to **Settings** → **Apps** → **Optional features**
- ▶ Click **Add a feature**
- ▶ Install **OpenSSH Client**
- ▶ Restart R / RStudio after installation

Note: SSH is an operating-system tool, not an R package.